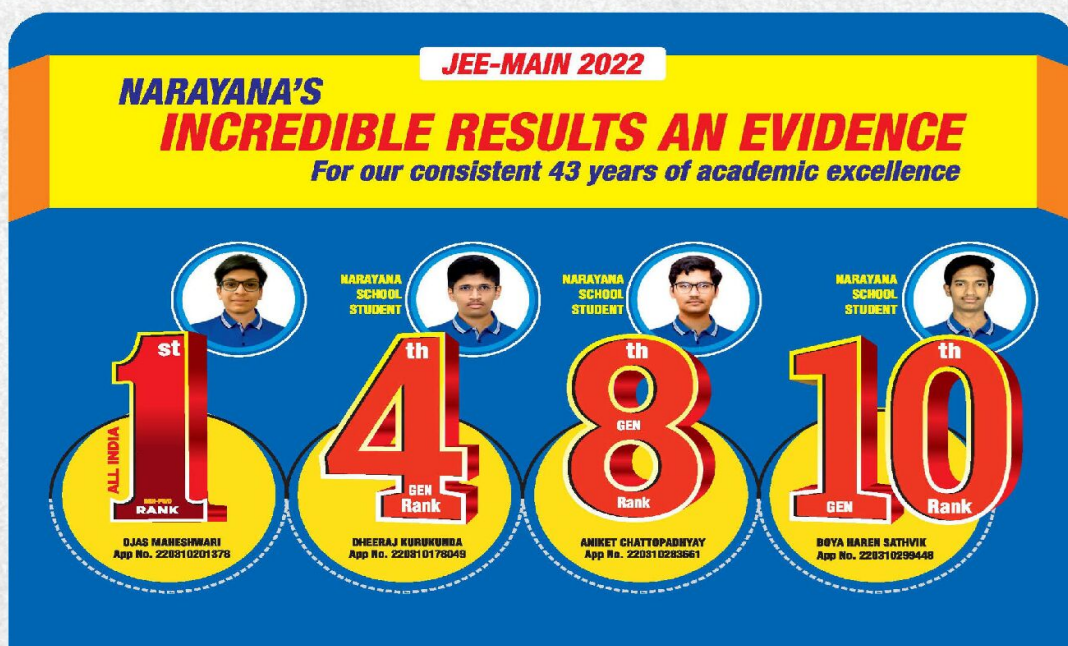


NARAYANA GRABS THE LION'S SHARE IN JEE-ADV.2022
EXCELLENCE IS OUR TRADITION 3402 RANKS OVERALL!



CHEMISTRY

PHYSICAL CHEMISTRY PART-I(JR)

JEE ADV IMPORTANT QUESTIONS

SECTION I: SINGLE OPTION CORRECT

1. The ionic product of water is 1×10^{-14} (mol/l)². The number of H⁺ ions that present in one millionthpart of 1 ml of pure water is -
 (A) 66.9 million (B) 52.2 million (C) 48.3 million (D) 60.3 million
2. Gas equation $PV = nRT$ is obeyed by:
 (A) only isothermal process (B) only adiabatic process
 (C) both (A) and (B) (D) none of these
3. [H⁺] and [OH⁻] ion in a 8 litre solution containing 9.25 gm/l of Ca(OH)₂ at 25°C, is
 (A) [OH⁻] = 0.25×10^{-7} , [H⁺] = 4×10^{-7} (B) [OH⁻] = 0.25, [H⁺] = 4×10^{-14}
 (C) [OH⁻] = 1.25×10^{-7} , [H⁺] = 4×10^{-7} (D) [OH⁻] = 2.5, [H⁺] = 4×10^{-14}
4. The degree of hydrolysis of a salt of weak acid and weak base in its 0.1 M solution is found to be 50%. If the molarity of the solution is 0.2 M, the percentage hydrolysis of the salt should be
 (A) 100% (B) 50% (C) 25% (D) none of these
5. Molar solubility of As₂S₃ in 0.3 M Al₂S₃ solution in terms of solubility product K_{sp} or As₂S₃ is—
 (A) $\sqrt{\frac{K_{sp}}{2.916}}$ (B) $3\sqrt{\frac{K_{sp}}{2.916}}$ (C) $\sqrt{\frac{K_{sp}}{1.2}}$ (D) $3\sqrt{\frac{K_{sp}}{1.2}}$
6. In a gaseous mixture at 20°C the partial pressure of the components are, :
 H₂: 150 Torr, CO₂: 200 Torr, CH₄: 300 Torr, C₂H₄: 100 Torr,
 Volume percent of H₂ is :
 (A) 26.67 (B) 73.33 (C) 80.00 (D) 20
7. The volume of water needed to dissolve 1 g of BaSO₄ (K_{sp} = 1.1×10^{-10}) at 25°C is –
 (A) 820 litre (B) 410 litre (C) 205 litre (D) none of these
8. The wavelength of a spectral line for an electronic transition is inversely related to:
 (A) Number of electrons undergoing transition
 (B) the nuclear charge of the atom
 (C) the velocity of an electron undergoing transition
 (D) the difference in the energy levels involved in the transition
9. Number of moles of electrons taken up when 1 mole of NO₃⁻ ions is reduced to 1 mole of NH₂OH is:
 (A) 2 (B) 4 (C) 5 (D) 6
10. Amount of oxalic acid present in a solution can be determined by its titration with KMnO₄ solution in the presence of H₂SO₄. The titration gives unsatisfactory result when carried out in the presence of HCl, because HCl:
 (A) furnishes H⁺ ions in addition to those from oxalic acid.
 (B) reduces permanganate to Mn²⁺.
 (C) oxidises oxalic acid to carbon dioxide and water.
 (D) gets oxidised by oxalic acid to chlorine.

11. Which of the following statements is not true?
 (A) The ratio of the mean speed to the rms speed is independent of the temperature.
 (B) The square of the mean speed of the molecules is equal to the mean squared speed at a certain temperature.
 (C) Mean kinetic energy of the gas molecules at any given temperature is independent of the mean speed.
 (D) The difference between rms speed and mean speed at any temperature for different gases diminishes as larger and yet larger molar masses are considered.
12. If velocity of an electron in I orbit of H atom is V, what will be the velocity of 3rd orbit of Li^{+2}
 (A) V (B) V/3 (C) 3 V (D) 9 V
13. When one gram mole of KMnO_4 reacts with HCl, the volume of chlorine liberated at NTP will be
 (A) 11.2 litres (B) 22.4 litres (C) 44.8 litres (D) 56.0 litres
14. A solution contains both Na_2CO_3 and NaHCO_3 was treated with excess of CaCl_2 solution and filtered. The precipitate weighed m_1 grams. On adding NaOH in drops to the filtrate avoiding excess, a further m_2 grams was precipitated. If after adding excess CaCl_2 , the solution (had not been filtered) but was simply boiled and then filtered, what would be the total weight of the precipitate
 (A) $(m_1 + m_2)$ grams (B) $(m_1 + m_2/2)$ grams
 (C) $(m_1 + m_2)/2$ grams (D) $(m_2 + m_1/2)$ grams
15. 0.7 g of a sample of $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$ were dissolved in water and the volume was made to 100 ml 20 ml of this solution required 19.8 ml of N/10 HCl for complete neutralization. The value of x is:
 (A) 7 (B) 3 (C) 2 (D) 5
16. Consider the following statements: The coefficient B in the virial equation of state

$$PV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

A: is independent of temperature

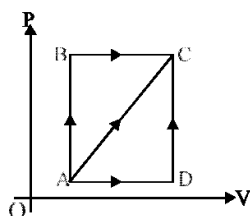
B: is equal to zero at Boyle temperature

C: has the dimension of molar volume

Which of the above statements are correct?

- (A) A and B (B) A and C (C) B and C (D) A, B and C
17. A thermodynamic process is shown in the following figure. The pressure and volumes corresponding to some point in the figure are:

$$p_A = 3 \times 10^4 \text{ pa}, p_B = 8 \times 10^4 \text{ pa}, V_A = 2 \times 10^{-3} \text{ m}^3, V_D = 5 \times 10^{-3} \text{ m}^3$$



In the process AB, 600 J of heat is added to the system and in BC, 200 J of heat is added to the system.

The change in internal energy of the system in the process AC would be:

- (A) 560 J (B) 800 J (C) 600 J (D) 640 J

18. Which of the following statements is correct in the context of $3d_{z^2}$ orbital?
 (A) This orbital consists of two positive lobes along the $\pm z$ axis and a negative doughnut in the xy plane.
 (B) This orbital consists of two negative lobes along the $\pm z$ axis and a positive doughnut in the xy plane.
 (C) This orbital consists of one negative lobe along the $\pm z$ axis and a negative doughnut in the xy plane.
 (D) This orbital consists of one positive lobe along the $\pm z$ axis and a negative doughnut in the xy plane.
19. 20 gm of N_2 at 300 K is compressed reversibly and adiabatically from 20 dm^3 to 10 dm^3 . Change in internal energy for the process is :
 (A) 284.8 J (B) 142.46 J (C) 1424.69 J (D) 3462.89 J
20. Which of the following is path function as well as an extensive property
 (A) temperature (B) internal energy
 (C) molar heat capacity (D) heat capacity
21. The shapes of PCl_4^+ , PCl_4^- and $AsCl_5$ are respectively:
 (A) square planar, tetrahedral and see-saw
 (B) tetrahedral, see-saw and trigonal bipyramidal
 (C) tetrahedral, square planar and pentagonal bipyramidal
 (D) trigonal bipyramidal, tetrahedral and square pyramidal
22. Number of electrons having $l + m$ value equal to zero in ${}_{24}\text{Cr}$ may be:
 (A) 14 (B) 10 (C) 13 (D) 12
23. If the enthalpy change for the reaction

$$\text{CH}_4(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow \text{CH}_3\text{Cl}(\text{g}) + \text{HCl}(\text{g})$$

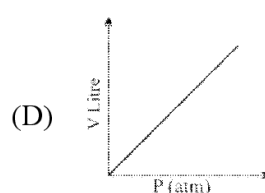
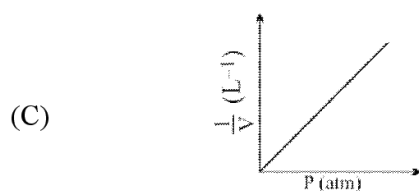
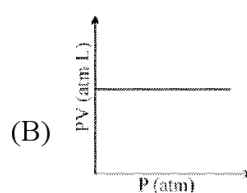
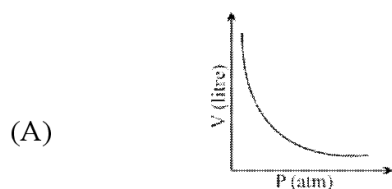
$\Delta H = -25 \text{ kcal}$. bond energy of C – H is 20 kcal mol^{-1} greater than the bond energy of C – Cl and bond energies of H – H and H–Cl are same in magnitude, then for the reaction :

- $\frac{1}{2}H_2(g) + \frac{1}{2}Cl_2(g) \rightarrow HCl(g); \quad \Delta H = ?$
- (A) -22.5 kcal/mol
 (B) -20.5 kcal/mol
 (C) -32.5 kcal/mol
 (D) -12.5 kcal/mol
- 24 The main factor responsible for weak acidic nature of B - F bond in BF_3 is;
 (A) Large electronegativity of F
 (B) Three centred two electron bond in BF_3
 (C) $P\pi - d\pi$ back bonding
 (D) $P\pi - P\pi$ back bonding
- 25 Among the following species, identify the isostructural pairs: $NF_3, NO_3^-, BF_3, H_3O^+, HN_3$
 (A) $[NF_3, NO_3^-]$ and $[BF_3, H_3O^+]$ (B) $[NF_3, HN_3]$ and $[NO_3^-, BF_3]$
 (C) $[NF_3, H_3O^+]$ and $[NO_3^-, BF_3]$ (D) $[NF_3, H_3O^+]$ and $[HN_3 - BF_3]$
- 26 Bond length of HCl is $1.275 \text{ \AA}$ ($q = 4.8 \times 10^{-10} \text{ e.s.u.}$) if $\mu = 1.02 \text{ D}$ then HCl is:
 (A) 100 % ionic (B) 83% covalent (C) 50% covalent (D) 40% ionic

SECTION II: MORE THAN ONE OPTION CORRECT

27. The dissociation of ammonium carbonate may be represented by the reaction
- $$NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2$$
- ΔH^0 for the forward reaction is negative. The equilibrium will shift from right to left if there is
- (A) a decrease in pressure
 (B) an increase in temperature
 (C) an increase in the concentration of ammonia
 (D) an increase in the concentration of carbondioxide
28. Which of the following samples of reducing agents is /are chemically equivalent to 25 ml of 0.2 N $KMnO_4$ to be reduced to Mn^{2+} and water.
- (A) 25 ml of 0.2 M $FeSO_4$ to be oxidized to Fe^{3+}
 (B) 50 ml of 0.1 M H_3AsO_3 to be oxidized to H_3AsO_4
 (C) 25 ml of 0.1 M H_2O_2 to be oxidized to H^+ and O_2
 (D) 25 ml of 0.1 M $SnCl_2$ to be oxidized to Sn^{4+}

- 29 The equilibrium of which of the following reaction will not be disturbed by the addition of an inert gas at constant volume
- (A) $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$ (B) $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$
- (C) $\text{CO}(\text{g}) + 2\text{H}_2(\text{g}) \rightleftharpoons \text{CH}_3\text{OH}(\text{g})$ (D) $\text{C}(\text{s}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + \text{H}_2(\text{g})$
- 30 1.2kg of iron pyrites (FeS_2) are roasted to convert sulphur in (FeS_2) to SO_2 with 80% efficiency. It is then oxidised to SO_3 which is then absorbed in water to give H_2SO_4 . If the maximum and minimum possible efficiencies are 60% and 30% respectively for any of these two steps, which of the following amounts of H_2SO_4 could possibly be obtained ? (Fe = 56, S = 32)
- (A) 0.8 mol (B) 1.5mol (C) 3.6mol (D) 5.7mol
- 31 Four gas balloons A, B, C, D of equal volumes containing H_2 , H_2O , CO , CO_2 respectively were pricked with needle and immersed in a tank containing CO_2 . Which of them will shrink after some time.
- (A) A (B) B (C) C (D) Both A and D
32. Heisenberg uncertainty principle is not valid for
- (A) moving electrons (B) motor car
- (C) stationary particles (D) all
33. When $\text{HCl}(\text{g})$ is passed through a saturated solution of common salt, pure NaCl is precipitated because
- (A) HCl is highly soluble in water
- (B) the ionic product $[\text{Na}^+][\text{Cl}^-]$ exceeds its solubility product (K_{sp})
- (C) the K_{sp} of NaCl is lowered by the presence of Cl^- ions
- (D) HCl causes precipitation
- 34 Which is true about an electron
- (A) Rest mass of electron is 9.1×10^{-28} gm
- (B) Mass of electron increases with the increase in velocity
- (C) Molar mass of electron is 5.48×10^{-4} gm/mole
- (D) e/m of electron is 1.7×10^8 coulomb/gm
- 35 Boyle's law is represented by -



37 According to Charles's law:

- (A) $V \propto \frac{1}{T}$ (B) $\left(\frac{dV}{dT}\right)_p = K$
 (C) $\left(\frac{dT}{dV}\right)_p = K$ (D) $\left(\frac{1}{T} - \frac{V}{T^2}\right) = 0$

36. The incorrect order of decreasing oxidation number of S in compounds is:

- (A) $\text{H}_2\text{S}_2\text{O}_7 > \text{Na}_2\text{S}_4\text{O}_6 > \text{Na}_2\text{S}_2\text{O}_3 > \text{S}_8$
 (B) $\text{H}_2\text{SO}_5 > \text{H}_2\text{SO}_3 > \text{SCl}_2 > \text{H}_2\text{S}$
 (C) $\text{SO}_3 > \text{SO}_2 > \text{H}_2\text{S} > \text{S}_8$
 (D) $\text{H}_2\text{SO}_4 > \text{SO}_2 > \text{H}_2\text{S} > \text{H}_2\text{S}_2\text{O}_8$

38. Which of the following are correct statements?

- (A) vander Waal's constant a is a measure of attractive force
 (B) vander Waal's constant b is also called covolume or excluded volume
 (C) b is expressed in L mol^{-1}
 (D) a is expressed in $\text{atm L}^2 \text{mol}^{-2}$

39. Which of the following will suppress the ionization of phthalic acid in an aqueous solution

- (A) KCl (B) H_2SO_4 (C) HNO_3 (D) NaOH

40 Following are the atoms having the number of neutrons and protons as given below:

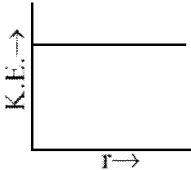
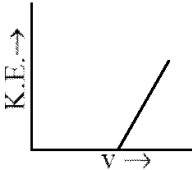
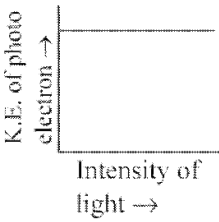
Atoms	Protons	Neutron
A	8	8
B	8	9
C	8	10
D	7	8
E	7	9

Select incorrect conclusion(s) :

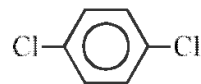
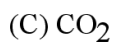
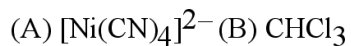
- (A) A, B and C, D are isotopes (B) A and D are isotones
 (C) A and E are isobars (D) A and B are isodiaphers

41 Three gases of densities A(0.82), B(0.25), C(0.51) are enclosed in a vessel of 4L capacity. Pick up the correct statement :

- (A) Gas A will tend to lie at the bottom
 (B) The number of atoms of various gases A, B, C are same
 (C) The gases will diffuse to form homogeneous mixture.
 (D) The average kinetic energy of each gas is same.

42. In which of the following reaction is $K_p < K_c$
- (A) $Co(g) + Cl_2(g) \rightleftharpoons CoCl_2(g)$
 (B) $Co(g) + 3H_2(g) \rightleftharpoons CH_4(g) + H_2O(g)$
 (C) $2BrCl(g) \rightleftharpoons Cl_2(g) + Br_2(g)$
 (D) $I_2(g) \rightleftharpoons 2I(g)$
43. Which of the following gases deviate fairly largely from ideal behaviour:
 (A) CO_2 (B) SO_2 (C) C_2H_6 (D) None
44. Which is correct graph (K.E. is kinetic energy of photo electron)
- (A) 
- (B) 
- (C) 
- (D) None of these
45. Which of the following are redox reactions?
- (A) $NaIO_3 + NaHSO_3 \rightarrow NaHSO_4 + Na_2SO_4 + I_2 + H_2O$
 (B) $FeCl_3 + K_4[Fe(CN)_6] \rightarrow KCl + Fe_4[Fe(CN)_6]_3$
 (C) $AgCl + Na_2S_2O_3 \rightarrow Na_3[Ag(S_2O_3)_2] + NaCl$
 (D) $NaBiO_3 + MnSO_4 + HNO_3 \rightarrow HMnO_4 + Bi(NO_3)_3 + NaNO_3 + Na_2SO_4 + H_2O$
46. The oxidation number of Cr = +6 in
 (A) $FeCr_2O_4$ (B) $KCrO_3Cl$ (C) CrO_5 (D) $[Cr(OH)_4]^-$
47. Which of the following have planar structure?
 (A) I_3^- (B) H_2O_2 (C) $Ni(CO)_4$ (D) $[Ni(CN)_4]^{2-}$
48. Which have odd-order bond ?
 (A) O_2^+ (B) O_2^- (C) NO (D) H_2^+
49. Which of the following mathematical relations are correct for an ideal gas?
 (A) $\left(\frac{\partial H}{\partial V}\right)_T = 0$ (B) $\left(\frac{\partial H}{\partial p}\right)_T = 0$ (C) $C_p - C_v > R$ (D) $\left(\frac{\partial C}{\partial V}\right)_T = 0$

50 Which has (have) zero value of dipole moment



51 The standard molar enthalpy of CO_2 is equal to

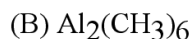
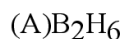
(A) the standard molar enthalpy of combustion of gaseous carbon

(B) the standard molar enthalpy of combustion of carbon (graphite)

(C) the sum of the standard molar enthalpies of formation of CO and O_2

(D) -394 kJ mol^{-1}

52 Three centre-two electron bonds exist in –



53 Which of the following statements are correct ?

(A) the entropy of an isolated system increases in an irreversible process

(B) the entropy of an isolated system remains unchanged in a reversible process

(C) Entropy can never decrease

(D) $\Delta S(\text{system})$ as well as (surroundings) are negative quantities

54. The factors that influence the heat of reaction are:

(A) the physical state of reactants and products

(B) the temperature

(C) the pressure or volume

(D) the method by which the final products are obtained

55 The condition for a spontaneity in a chemical reaction is

(A) $(\Delta G)_{T,P} \leq 0$

(B) $(\Delta U)_{S,V} \leq 0$

(C) $(\Delta H)_{S,P} \leq 0$

(D) $(\Delta S)_{U,V} \geq 0$

SECTION III: COMPREHENSIONS

COMPREHENSION # 1

Paragraph for Questions Nos. 56 to 58

In the quantum mechanics there is operator ($\hat{A}$) for every physically observable property like momentum (linear or angular), position K.E., total energy etc., and whenever the function (ψ) is operated upon by the operator of any property „ we get back ψ multiplied by a constant (a) which is the value of the property under consideration. The function ψ obeying the condition is called eigenfunction of the operator $\hat{A}$ and the constant “a” is called eigen-value. Mathematically

$$\hat{A} \psi = a \cdot \psi$$

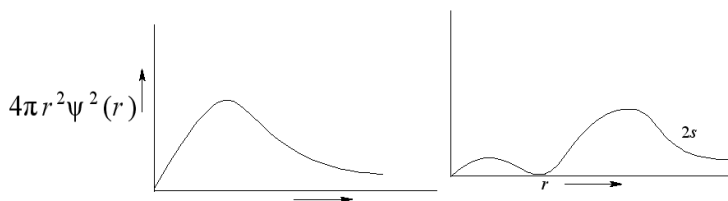
This equation is called eigen-value equation.

The Schrodinger wave equation is the energy eigen value equation. The operator for total energy is called Hamiltonian operator (H) defined as

$$H = -\frac{h^2}{8\pi^2 m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) + V$$

Where V = PE, m = mass of microparticle.

Solving Schrodinger wave equation for H-atom after putting proper value of V, we get wave functions (ψ) and corresponding energy values. Each wave function denotes an orbital. ψ^2 measures the probability of finding the electron and $\psi^2 dx$ gives the probability of finding the electron in the region x and x + dx. $4\pi r^2 \psi^2(r) dr$ gives the probability of finding the electron in spherical shell of thickness dr at a distance r from the nucleus. A plot of radial distribution function $[4\pi r^2 \psi^2(r)]$ vs. distance from the nucleus (r) gives the radial probability distribution curve of electron. For 1s and 2s orbitals each containing only one electron, these curves are as given below



The wave functions of 1s and 2s orbitals of hydrogenic atom or ion are given below:

$$\psi_{1s} = 2 \left(\frac{Z}{a_0} \right)^{1/2} e^{-Zr/a_0}$$

$$\psi_{2s} = \left(\frac{Z}{2a_0} \right)^{1/2} \left(2 - \frac{Zr}{a_0} \right) e^{-Zr/2a_0}$$

In 2s-orbital there occurs a point at the radius at which curve touches X – axis. Here the probability of Finding the electron is zero. This is called radial node.

56. Which of the functions given below is not the eigen function of the operator $\frac{d^2}{dx^2}$?

- (A) e^{-3x} (B) $\sin 4x$ (C) $4x^2$ (D) $\cos 2x$

57. An electron moves along x-axis and restricted to move only between $x = 0$ and $x = L$. Hence

(A) $\int_0^L \psi^2 dx = 1$ (B) $\int_0^L \psi^2 dx = 0$

(C) $\int_0^L \psi^2 dx > 0$ (D) $\int_0^L \psi^2 dx < 1$

58. An electron is moving along X-axis in absence of any external field. The energy eigen-value of electron with $\psi = e^{-5x}$, is

(A) $\frac{25}{8} \frac{h^2}{\pi^2 m}$ (B) $\frac{3}{8} \frac{h^2}{\pi^2 m}$

(C) $\frac{h^2}{8\pi^2 m}$ (D) $\frac{h^2}{4\pi^2 m}$

COMPREHENSION # 2

Paragraph for Questions Nos. 59 to 60

From the Bohr's model and De-broglie's theory it can be proved that number of waves in an orbit is equal to its principal quantum nos.(n). The orbital motion of the electron around the nucleus is described by angular momentum of the electron which is characterised by azimuthal quantum number, l . l have all possible whole number values from 0 to $n - 1$ and magnitude of angular

momentum is given by $(\ell / \ell + 1)^{1/2} \hbar$. The various sub shells are distributed as s, p, d, f,...according to the value of $l = 0, 1, 2, 3, \dots$ respectively.

Angular momentum about the z-axis is quantised and given by z-component of angular momentum

$= m_l \cdot \cos \hbar \quad m = \ell(\ell + 1)^{1/2} \cos \theta$ and is known as magnetic quantum number. Possible values of m range from $-l$ through 0 to $+l$, thus making a total of $2l + 1$ values. The possible values of m indicates the number of degenerate orbitals in a given energy sub level within a principal energy level.

59. When the value of $l = 1$, then the values of θ are

(A) $60^\circ, 120^\circ, 180^\circ$

(B) $0, 90^\circ, 180^\circ$

(C) $45^\circ, 90^\circ, 135^\circ$

(D) $30^\circ, 60^\circ, 90^\circ$

60. The difference between orbital angular momentum of a 2s and a 3d electron is

(A) $\sqrt{2}\hbar$

(B) 0

(C) $\sqrt{4}\hbar$

(D) $\sqrt{6}\hbar$

COMPREHENSION # 3

Paragraph for Questions Nos. 61 to 63

The radial probability at a distance r from the nucleus is given by the function $4\pi r^2 \psi^2 dr$. In this function, as the probability factor ψ^2 decreases, the volume factor $4\pi r^2 dr$ increases with increase in value of r . The probability increases gradually as the distance increases, goes to a maximum and then begins to decrease. For 1s orbital of hydrogen atom, the maximum value is found by differentiating radial probability distribution with respect to r and setting the derivative equal to zero as

$$\frac{dp(r)}{dr} = \frac{4}{a_0^3} \left[-\frac{2r^2}{a_0} + 2r \right] e^{-2e/a_0} = 0$$

The radial motion of an electron in an atom is controlled by the centrifugal force $\ell(\ell + 1)/r^3$ as well as by coulombing force of attraction $(-Ze^2/r^2)$ or

$$\frac{\ell(\ell + 1)}{r^3} - \frac{Ze}{r^2} = 0$$

61. The probability of radial distribution function of an electron is zero becomes possible when distance r is equal to

- (A) $0.529\text{\AA}$ (B) zero (C) $2 \times 0.529\text{\AA}$ (D) $\frac{0.529}{2}\text{\AA}$

- 62 In two different cases under similar conditions, an electron tends to stay farther from the nucleus in case I with respect to case II. It shows that
 (A) Orbital angular momentum in, case I > case II
 (B) Orbital angular momentum in, case I < case II
 (C) Orbital angular momentum in, case I = case II
 (D) None of the above
- 63 The electron distribution in 1s orbital is found to be symmetrical. Considering θ and ϕ be the angles involved in polar coordinate, it may be concluded that:
 (A) ψ^2 is dependent on θ (B) is dependent on ϕ
 (C) is dependent on both θ and ϕ (D) is independent of θ and ϕ

COMPREHENSION # 4

Paragraph for Questions Nos. 64 to 67

Dry air contains 80 % N_2 and 20 % O_2 by volume. Calculate density of dry air at 27°C and 1 atm pressure. If relative humidity of moist air is 50 % and V.P. water at 27°C is 40mm. Calculate the density (g/L) of moist air also.

Assume ideal behavior of each gas. Relative humidity is defined as the ratio of the

partial pressure of moisture in air to its V.P. Three students "X", "Y" and "Z" solved 1st part of the problem in the following three different ways:

Solution of "X": One litre dry air contains 0.8 L N_2 and 0.2 L O_2 at the partial pressure of 0.8 atm and 0.2 atm respectively while each at 27°C i.e. 300K. He then calculated number of mole (n) of each gas using the ideal gas equation $PV = nRT$. He then multiplied number of mole with respective MW to get the mass of each gas in gram. Adding the masses of the two gases he was able to find the mass of 1 litre air in gram and the same he reported as the density of dry air.

Solution of "Y": One litre dry air contains 0.8 L N_2 and 0.2 L O_2 each at a pressure of 1.0 atm and temperature of 27°C i.e. 300K. Thereafter he followed the same method as adopted by "X" and reported the density of dry air.

Solution of "Z": One litre dry air may be supposed to contain 1.0 L each of N_2 and O_2 at 27°C but at a partial pressure of 0.8 atm and 0.2 atm, respectively. Then he followed the method as adopted by his aforesaid guys, and reported the density of dry air.

64. The correct solution(s) is (are) that of:
 (A) both X and Y (B) both Y and Z
 (C) All the three (D) Only X
65. If solution "X" is wrong what conceptual mistake he has committed:
 (A) Boyle's law is violated (B) Charle's law is violated
 (C) Graham's law is violated (D) There is no violation of any gas law
66. If solution "Y" is wrong what conceptual mistake he has committed:
 (A) Dalton's law of partial pressure is violated
 (B) Avogadro's hypothesis has been violated
 (C) Boyle's law is violated
 (D) There is no violation of any gas law
67. Solution of "Z" is wrong because
 (A) Boyle's law is violated (B) Charle's law is violated
 (C) Avogadro's hypothesis is violated (D) The solution is correct

COMPREHENSION # 5

Paragraph for Questions Nos. 68 to 70

Compressibility factor (Z): Real gases deviate from ideal behaviour due to the following two faulty assumptions of kinetic theory of gases.

- i) Actual volume occupied by the gas molecule is negligible as compared to the total volume of the gases.
- ii) Forces of attraction and repulsion among the gas molecules are negligible.

To explain, the extent of deviation of the real gas from ideal behaviour in terms of compressibility or, compression factor (Z), which is function of pressure and temperature for real gases.

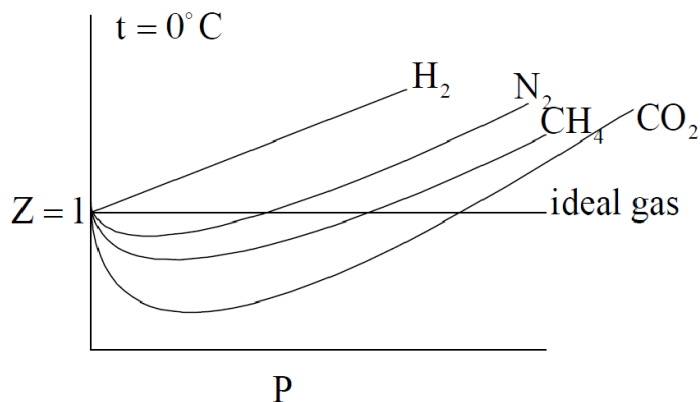
$$Z = \frac{P_0 V_0}{nRT}$$

For ideal gases $Z = 1$

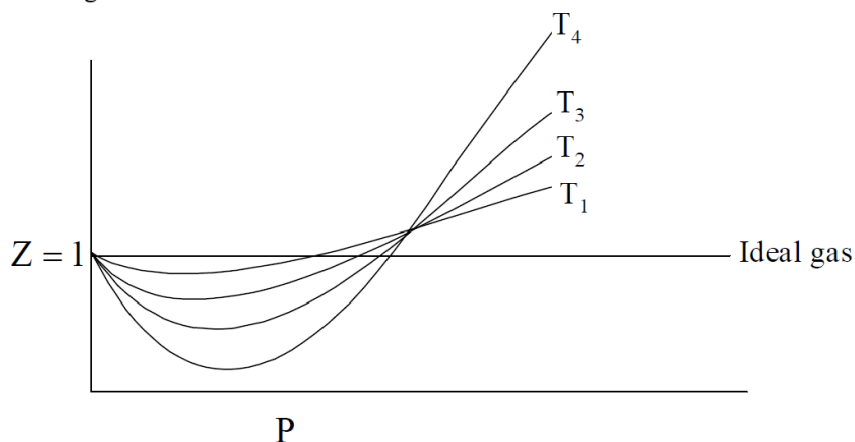
For real gases either $Z > 1$ or $Z < 1$

When $Z > 1$, then it is less compressible because force of repulsion dominates over force of attraction when $Z > 1$, force of attraction dominates over the force repulsion and it is more compressible.

Graph in between Z & P is shown as under



On increasing temperature, Z increases and approaches to unity. Graph between Z and P at different temperature for the same gases are as under:



68. Which of the following statements are correct?
- (A) The compressibility factor for ideal gases dependent on temperature and pressure.
- (B) $\left(\frac{dZ}{dP}\right)_r$ for real gas is independent of pressure.
- (C) for all real gases have same value
- (D) for different real gases have different values.
69. Which of the following is the correct order of temperature shown in the above graph Z vs P for the same gases.
- (A) $T_4 > T_3 > T_2 > T_1$
- (B) $T_1 > T_2 > T_3 > T_4$
- (C) $T_1 > T_2 > T_4 > T_3$
- (D) $T_3 > T_4 > T_2 > T_1$
70. Which of the following statements is correct for gas A having molar mass 16 g and density 0.75 g/litre at 2 atmospheric pressure and 27°C temperature?
- (A) Force of attraction are dominating than force of repulsion among the gas molecules.
- (B) Force of repulsion are dominating than force of attraction among the gas molecules.
- (C) Gas molecules are behaving ideally
- (D) None

COMPREHENSION # 6

Paragraph for Questions Nos. 71 to 73

The van der waal's equation of state for 1 mole real gas is ... (i)

The virial equation for 1 mole real gas is as under:

Where A, B and C are constants which are known as 2nd, third and fourth virial coefficients respectively.

Since we know that $V = f(P, T, n]$

Since ideal gas equation for 1 mole ideal gas is $PV = RT$... (ii)

The temperature at which real gas obey ideal gas equation is known as Boyle's temperature i.e. T_B .

- 71 If equation (ii) be one of the form of equation (i), then value of "A" will be

(A) $b - \frac{a}{RT}$

(B) $a - \frac{b}{RT}$

(C) $RT - \frac{a}{b}$

(D) $\frac{RT \times b}{a}$

- 72 Which of the following is the correct statement about the Boyle's temperature (T_B)?

(A) Temperature at which second virial coefficient becomes zero

(B) Temperature at which first virial coefficient becomes zero

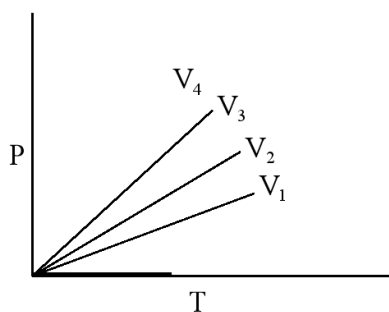
(C) The value of T_B is equal to $\frac{a}{Rb}$

(D) Both (a) and (c) are correct

- 73 The ideal gas equation for 1 mole of ideal gas is given as under

$$PV = nRT$$

The graph is drawn in between P and T for different isochoric process, or different constant volume, which gives different straight lines shown as under:



then which of the following is the correct order of volume?

(A) $V_4 > V_3 > V_2 > V_1$

(B) $V_1 > V_2 > V_3 > V_4$

(C) $V_1 > V_2 > V_4 > V_3$

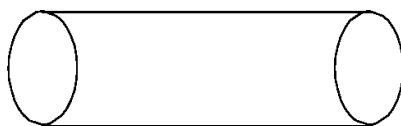
(D) $V_2 > V_1 > V_3 > V_4$

COMPREHENSION # 7
Paragraph for Questions Nos. 74 to 76

Diffusion is the one of the most striking property of the gas, due to which one gas intermixes with other against gravity at low pressure gradient at constant temperature. And according to Graham, rate of diffusion of a diffusing gas is inversely proportional to the square root of the density or molecular mass of the diffusing gas i.e.,

$$r \propto \frac{1}{\sqrt{d}} \propto \frac{1}{\sqrt{m}} \quad \dots(i)$$

The rate of diffusion inside an open tube is directly proportional to the length



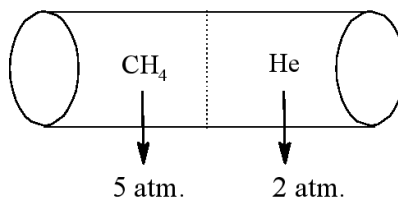
$$r = \frac{v}{t} = \left(\frac{A}{t} \right) \times \text{length}$$

$$r \propto \text{length}$$

But, effusion is a special kind of diffusion, possible at high pressure gradient only. It is a kind of forced diffusion. Infact incase of effusion rate of intermixing will be influenced by pressure gradient. So, in effusion pressure must be considered i.e.,

$$r \propto \frac{P}{\sqrt{m}} \propto \frac{P}{\sqrt{d}}$$

- 74 At the start of an experiment one end of a U-tube of 6 mm glass tubing is immersed in conc. NH_3 solution and the other end is immersed in conc. HCl solution. At the point in the tube where vapours of NH_3 and HCl meet, a white cloud of NH_4Cl (s) forms. At what fraction of the distance along the tube from the NH_3 solution does the white cloud first form?
- (A) 0.068 (B) 6.82 (C) 0.422 (D) 4.22
- 75 A long cylindrical glass tube, equipped with a process disc at the centre contain CH_4 gas at 5 atm. on one side and He gas at 2 atm on the other side of the disc as shown in the diagram.



Disc is permeable to both gases and rate of diffusion is directly proportional to the gas pressure and inversely proportional to square root of molar masses as;

$$- \frac{dP}{dt} = \frac{kP}{\sqrt{M}}$$

If k for the CH_4 is $2.5 \times 10^{-2} \text{ sec}^{-1}$, determine time after which pressure of CH_4 chamber will drop to 4 atm?

- (A) 8.778 sec. (B) 87.88 sec. (C) 48.2 sec. (D) 40 sec.

- 76 If a space capsule is filled with Ne gas at 1 atm and 290K. The gas effuses through a pin-hole into outer space at such a rate that the pressure drops by 0.30 torr/s. If the capsule were filled with NH_3 at the same temperature and pressure what would be the rate of pressure drop?

- (A) 0.33 torr/s (B) 0.20 torr/s (C) 0.13 torr/s (D) 3.3 torr/s

COMPREHENSION # 8

Paragraph for Questions Nos. 77 to 79

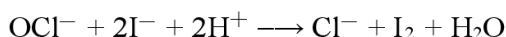
“Percentage of available chlorine in the sample of bleaching powder”

Bleaching powder is a mixture of calcium hypochlorite (CaOCl_2) and the basic chloride $\text{CaCl}_2 \cdot \text{Ca}(\text{OH})_2 \cdot \text{H}_2\text{O}$. Free slaked lime is also present in addition.

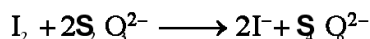
Available chlorine in bleaching powder is the amount of chlorine liberated by the action of dilute acids on bleaching powder according to the equation. $\text{OCl}^- +$



The available chlorine is expressed as the percentage by weight of the bleaching powder. It is measured by iodometric titration. The bleaching powder solution or suspension is treated with an excess of KI and then acidified with acetic acid where I_2 is liberated



The liberated I_2 is determined with standard $\text{Na}_2\text{S}_2\text{O}_3$ solution using starch as indicator near the endpoint



- 77 0.05 moles of CaOCl_2 reacts with excess of KI and liberated I_2 is titrated with V ml of 0.1M hypo solution then value of V is

- (A) 500 ml (B) 1000 ml (C) 2000 ml (D) 250 ml

- 78 A 20 ml (specific gravity 1.02) of chlorine water is treated with excess of KI and the liberated Iodine required 25 ml of 0.2 N $\text{Na}_2\text{S}_2\text{O}_3$. The percentage of free chlorine in chlorine water is

- (A) 0.87 (B) 87 (C) 0.78 (D) 0.82

- 79 The volume of 0.2 M $\text{Na}_2\text{S}_2\text{O}_3$ equivalent to I_2 liberated by the oxidation of excess KI by 10 millimole of bleaching powder (pure) is

- (A) 50 ml (B) 20 ml (C) 25 ml (D) 100 ml

COMPREHENSION # 9

Paragraph for Questions Nos. 80 to 83

Like acid base titration, in redox titration also, the equivalence point is reached when the reducing agent is completely oxidized by the oxidizing agent. But contrary to the acid-base titrations, oxidizing agents can themselves be used as internal indicator in redox titration e.g. $\text{Cr}_2\text{O}_7^{2-}$ (orange yellow), Cr^{3+} (green), MnO_4^- (purple), Mn^{2+} (light pink), where strength of the solution may be expressed as molarity i.e. number of moles of solute per litre of solution.

- 80 In a titration experiment, a student finds that 23.48 ml of a NaOH solution are needed to neutralize 0.5468g of KHP (molecular formula $\text{KHC}_8\text{H}_4\text{O}_4$). What is the concentration in molarity of NaOH solution?
(A) 0.114 M (B) 0.228M (C) 0.057M (D) 0.028 M
- 81 A 16.42 ml volume of 0.1327M KMnO_4 solution is needed to oxidize 25.00 ml of a FeSO_4 solution in an acidic medium. What is the number of moles of FeSO_4 being oxidized for the reaction
$$5\text{Fe}^{2+} + \text{MnO}_4^- + 8\text{H}^+ \longrightarrow \text{Mn}^{2+} + 5\text{Fe}^{3+} + 4\text{H}_2\text{O}$$

(A) 2.18×10^{-2} ml (B) 1.09×10^{-2} ml
(C) 0.545×10^{-2} ml (D) 0.272×10^{-2} ml
- 82 A purple coloured solution is added from a burette to FeSO_4 solution kept in the flask. After sometime, the purple colour changes to light pink. The ion formed from that solution is
(A) MnO_4^- (B) Fe^{2+} (C) Fe^{3+} (D) Mn^{2+}
- 83 Concentrated aqueous sulphuric acid is 98% H_2SO_4 by mass and has a density of 1.84 g/ml. What volume of the concentrated acid is required to make 5 litre of 0.50M H_2S_4 solution
(A) 271.7 ml (B) 13.5 ml (C) 135.85 ml (D) 27.1 ml

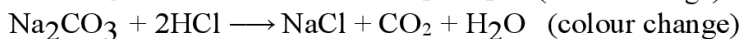
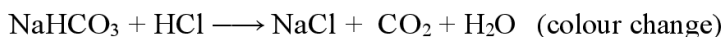
COMPREHENSION # 10

Paragraph for Questions Nos. 84 to 86

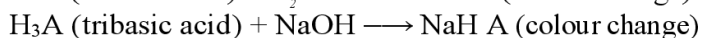
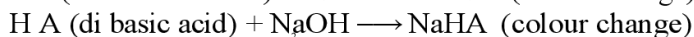
Double Titration

1. Methyl orange as an indicator

The volume of 0.1 N HCl used in the titration will correspond the neutralization as directed.

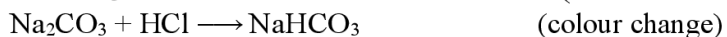


The volume of 0.1M NaOH in the titration will neutralize as directed



2. **Phenolphthelien as an indicator**

The volume of 0.1 N HCl used in the titration will correspond the neutralization as directed.



The volume of 0.1M NaOH in the titration will neutralize as directed



- 84 0.01 mole of H_3PO_4 and 0.01 mole H_3PO_2 reacts with V ml standard NaOH in presence of phenolphthelien indicator volume of NaOH used is
(A) 500 ml (B) 400 ml (C) 300 ml (D) 200 ml
- 85 20 ml of x M HCl neutralizes completely 10 ml of 0.1 M solution and a further 5 ml of 0.2 M solution to methyl orange end point. The value of x is:
(A) 0.167 M (B) 0.133M (C) 0.15 M (D) 0.2 M
- 86 Certain moles of is dissolved in excess of NaOH. The resulting solution is divided into two equal parts. One part needs 30 ml of 2.5 N HCl and other part needs 40 ml of 2.5 N HCl using phenolphthalein and methyl orange indicator respectively. The mass of dissolved is:
(A) 2.5×10^{-2} (B) 5×10^{-2} (C) 10×10^{-2} (D) 1.5×10^{-2}

COMPREHENSION # 11

Paragraph for Questions Nos. 87 to 89

Bond length is the average distance between the nuclei of the two atoms held by a bond. This represents the internuclear distance corresponding to minimum potential energy for the system. Main factors which affect the bond length are given below:—

- Multiple bonds are shorter than corresponding single bonds
- Sometimes single bond distances are some what larger than double of their respective covalent radii (e.g. F_2). It is due to strong repulsive interaction between the lone-pair electrons on adjacent atoms.
- Some times single bond distances are some what shorter than double of their respective covalent radii because bonds acquire some partial double bond character. This normally happens when one atom having vacant orbital and another atom containing lone pair. It is also possible it become shorter due to high ionic character in the covalent bond.

87. Which is not true about the N – N bond length among the following species

- I. $\text{H}_2\text{N} - \text{NH}_2$ II. N_2
 III. $\text{HN}^+ - \text{NH}_2$ IV. N_2O

- (A) N — N bond length is shortest in II
 (B) N — N bond length in I is shorter than that of in III
 (C) N — N bond length in III is shorter than that of in I
 (D) N — N bond length IV is intermediate between I and III

88. The correct order about of C—O bond length is

- I. CO, II. CO_2 III. CO_3^{2-}

- (A) $\text{II} < \text{I} < \text{III}$ (B) $\text{III} < \text{II} < \text{I}$ (C) $\text{I} < \text{II} < \text{III}$ (D) $\text{I} < \text{III} < \text{II}$

89. The bond angle in Cl_2O is more than that in F_2O because of so many reasons like

- (a) $p\pi - d\pi$ overlap in Cl_2O which is not possible in F_2O
 (b) There is no lp – lp repulsion in Cl_2O which is present in F_2O
 (c) F is more electronegative than oxygen while Cl is less electronegative than oxygen

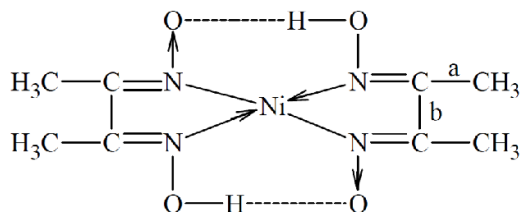
The correct reason is (are)

- (A) both a and b (B) only a (C) only b (D) all the three

COMPREHENSION # 12

Paragraph for Questions Nos. 90 to 92

“Dimethyl glyoxime is an important reagent, which is used in the detection of Ni^{2+} ion in fourth group of qualitative analysis and also in its gravimetric estimation. The compound formed by Ni^{2+} ions with dimethyl glyoxime is diamagnetic with the structure as given below.



‘a’ and ‘b’ are C—C bond length”.

90. Which of the following is correct regarding the bond length ‘a’ and ‘b’.

- (A) $a = b$ (B) $a > b$ (C) $b > a$ (D) none of the above

91. The necessary conditions for the formation of intramolecular hydrogen bonding is

- (A) The ring formed as a result of hydrogen bond should be planar
 (B) A five or six membered ring should be formed
 (C) Interacting atoms should be placed in such a way that there is minimum strain during the ring formation
 (D) All of these

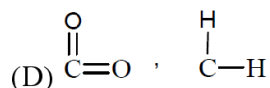
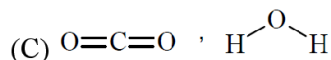
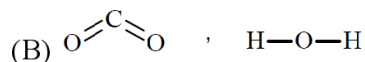
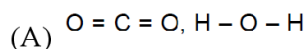
- 92 The atomic orbitals which the central atom is using in hybridisation are:
 (A) $4s, 4p_x, 4p_y$ and $4p_z$ (B) $3d^2, 4s, p_x$ and $4p_y$
 (C) $3d_{x^2-y^2}, 4s, 4p_x$ and $4p_y$ (D) $3d_{x^2-y^2}, 4s, 4p_y$ and $4p_z$

COMPREHENSION # 13

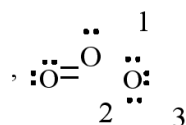
Paragraph for Questions Nos. 93 to 95

The formal charge is the difference between the number of valence electrons in an isolated (i.e. free) atom and the number of electrons assigned to that atom in a Lewis structure. While as a result of polarization, the molecule possesses a dipole moment which is the product of charge on the dipole and distance between them. For a molecule, the net dipole moment is the vector addition of all the dipole moment. Just as all the covalent bonds have some partial ionic character the ionic bonds also have partial covalent character. The polarizing power of the cation, the polarisability of the anion and the extent of distortion of anion are the factors, which determine the percent covalent character of the ionic bond.

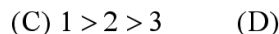
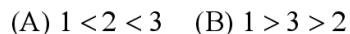
- 93 Experiment shows that H_2O has a dipole moment whereas CO_2 has not. Point out the structures which best illustrate these factor.



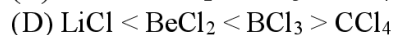
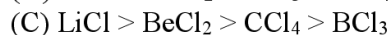
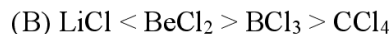
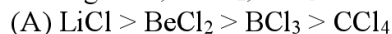
- 94 It has been seen that in O_3 , the central O atom is bonded to two other oxygen atoms as



Arrange oxygen atoms marked as 1, 2 and 3 in order of their decreasing formal charge:



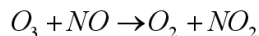
- 95 Among $LiCl$, $BeCl_2$, BCl_3 and CCl_4 , covalent bond character follows the order:



COMPREHENSION # 14

Paragraph for Questions Nos. 96 to 98

The depletion of ozone in the stratosphere has been a matter of great concern among scientists in recent years. It is believed that ozone can react with nitric oxide (NO) that is discharged from the high altitude jet plane. The reaction is:



- 96 If 0.740 g of O_3 reacts with 0.670 g of NO, how many gram of NO_2 will be produced?
(A) 0.71 g (B) 0.74 g (C) 0.68 g (D) 0.81 g
- 97 Which compound is the limiting reactant?
(A) NO (B) O_3
(C) Both are in equimolar ratio (D) Both are in stoichiometric ratio
- 98 Number of moles of the excess reactant remaining at the end of the reaction is
(A) 0.007 mol O_3 (B) 0.014 mol O_3
(C) 0.007 mol NO (D) 0.014 mol NO

COMPREHENSION # 15

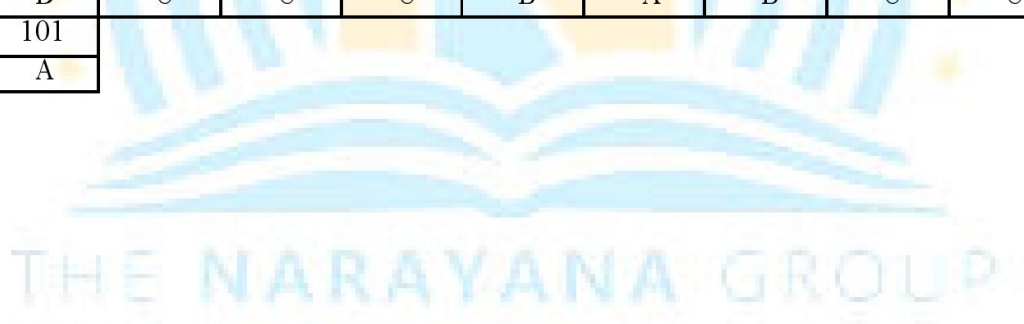
Paragraph for Questions Nos. 99 to 101

8.7 gm of a sample of is used in a chemical reaction with HCl to release. Released is used to displace from excess of KI solution. The iodine hence released is estimated using hypo and consumes 100 ml of hypo solution. 12.5 ml of same hypo solution is required for complete reaction with 25 ml of 0.5 N solution.

- 99 % purity of is
(A) 4.35 (B) 43.5 (C) 50 (D) 6.25
- 100 Wt. of which may be completely neutralized by the amount of HCl made to react quantitatively with pure in the sample is
(A) 106 g (B) 10.6 g (C) 5.3 g (D) 53 g
- 101 Number of milli moles of iodine in the solution used with hypo are
(A) 6.25 (B) 12.5 (C) 10 (D) 20

ANSWER KEY

Q.No.	1	2	3	4	s	6	7	g	9	io
Ans.	D	C	B	B	A	D	B	D	D	B
Q.No.	11	12	13	14	15	16	17	18	19	20
Ans.	B	A	D	D	C	C	A	A	C	D
Q.No.	21	22	23	24	25	26	27	28	29	30
Ans.	B	D	A	D	C	B	ABC	AD	ABC D	BCD
Q.No.	31	32	33	34	35	36	37	38	39	40
Ans.	AB C	BC	BD	AB CD	ABC	CD	BCD	ABC D	BC	AD
Q.No.	41	42	43	44	45	46	47	48	49	50
Ans.	CD	AB	AB C	BC	AD	BC	AD	ABC D	AB D	ACD
Q.No.	51	52	53	54	55	56	57	58	59	60
Ans.	BD	AB	AB C	AB C	ABC D	C	A	A	C	D
Q.No.	61	62	63	64	65	66	67	68	69	70
Ans.	B	A	D	B	A	D	D	D	A	B
Q.No.	71	72	73	74	75	76	77	78	79	80
Ans.	A	D	B	A	B	A	B	A	D	A
Q.No.	81	82	83	84	85	86	87	88	89	90
Ans.	B	D	C	C	C	B	C	C	D	B
Q.No.	91	92	93	94	95	96	97	98	99	100
Ans.	D	C	C	C	B	A	B	C	C	B
Q.No.	101									
Ans.	A									





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EXCELLENCE IS OUR TRADITION

3402 RANKS OVERALL!

5 RANKS IN OPEN CATEGORY ONLY FROM NARAYANA IN TOP 10 AIR



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JEE MAIN -2022

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INCREDIBLE RESULTS AN EVIDENCE*For our consistent 43 years of academic excellence*All India Top Ranks Below 10 *all categories*ALL CATEGORY
BELOW 150
RANKS

151

ALL CATEGORY
BELOW 1000
RANKS

612